

CYBERSECURITY PORTFOLIO

HANDS-ON LAB · LINUX SECURITY

Linux Security Assessment & Hardening

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Portfolio Edition

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Original technical content and evidence preserved.
Other participant names removed for public portfolio use.

Securing Linux Environments Against Common Attacks

Week 1, Day 3-5

8/5/2026-8/7/2026

Day 3 Lab

Step 1:

SSH into your assigned target VM using the credentials provided.

```
PS C:\Users\cjyan> ssh student@52.249.216.214
The authenticity of host '52.249.216.214 (52.249.216.214)' can't be established.
ED25519 key fingerprint is SHA256:kLgZBQ0z4tJkCziWLL0UYRgi0xu3n2WdsUoyF+jjngg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.249.216.214' (ED25519) to the list of known hosts.
student@52.249.216.214's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1021-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Aug  5 19:23:28 UTC 2026

System load:  0.08               Processes:            130
Usage of /:   5.9% of 28.02GB    Users logged in:     1
Memory usage: 4%                IPv4 address for eth0: 172.16.0.4
Swap usage:   0%

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.

Last login: Wed Aug  5 19:23:29 2026 from 40.71.199.151
$
```

Shows SSH into the victim target.

Step 2:

Explore the top-level directory structure: `ls -la /` and record what each major directory (`/etc`, `/var`, `/home`, `/usr`, `/tmp`) is used for.

```
$ ls -la
total 24
drwxr-x--- 3 student student 4096 Aug  5 16:55 .
drwxr-xr-x 5 root    root    4096 Aug  5 16:56 ..
-rw-r--r-- 1 student student 220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 student student 3771 Mar 31  2024 .bashrc
drwx----- 2 student student 4096 Aug  5 16:55 .cache
-rw-r--r-- 1 student student 807 Mar 31  2024 .profile
$
```

In the screenshot on top we were looking to see what the directory structure looked like.

We found `/etc` and found that it is for system wide configurations

```
$ ls /etc
ModemManager      cifs-utils      depmod.d      gss             ld.so.conf      magic           newt            profile         rsyslog.d      sudo.conf      update-notifier
PackageKit         cloud           dhcpcd        hdparm.conf    ld.so.conf.d    magic.mime      nftables.conf  profile.d      screencrc     sudo_logsrvd.conf  usb_modeswitch.conf
X11               console-setup  dhcpcd.conf   host.conf      ldap            manpath.config nsswitch.conf  protocols     security      sudoers.d      usb_modeswitch.d
adduser.conf      credstore      dpkg          hostname       legal           mdadm          nvme           python3        selinux       sudoers.d      vconsole.conf
alternatives      credstore.encrypted  e2scrub.conf  hosts          libaudit.conf  mime.types     opt            python3.12     sensors.d     supercat       vim
apparmor          cron.d         ec2_version   hosts.allow    libblockdev     mke2fs.conf    os-release     rc0.d          sensors3.conf  sysctl.conf     vmware-tools
apparmor.d        cron.daily     environment   hosts.deny     libbverbs.d     modprobe.d     overlayroot.conf  rc1.d          services      sysctl.d       vtrgb
appport           cron.hourly    ethertypes    init           libl3           modules        overlayroot.local.conf  rc2.d          sgml          sysstat        waagent.conf
apt              cron.monthly   fstab         init.d         locale.alias    modules-load.d  pam.conf       rc3.d          shadow        systemd        wgetrc
bash.bashrc       cron.weekly    fuse.conf     initramfs-tools  locale.conf     mtab           pam.d          rc4.d          shadow-       xattr.conf
bash_completion   cron.yearly    fwupd        inputrc        locale.gen      multipath       passwd         rc5.d          shells        timezone       xdg
bash_completion.d  crontab       gai.conf     iproute2       localtime       multipath.conf  passwd-        rc6.d          skel          tmpfiles.d     xml
bindresvport.blacklist  cryptsetup-initramfs  gnutils       iscsi          logcheck        nanorc          perl           rc5.d          sos           ubuntu-advantage  zsh_command_not_found
binfmt.d          crypttab       groff        issue          login.defs       needrestart     pki            request-key.conf  ssh           ucf.conf
byobu            dbus-1        group        issue.net      logrotate.conf  netconfig      plymouth       request-key.d    ssl           udev
```

We found `/var` and that it shows data that is variable

```
$ ls /var
backups cache crash lib local lock log mail opt run snap spool tmp
$
```

We found `/home` and that it shows home directories

```
$ ls /home
azureuser oldservice student
```

We found `/usr` it shows application and libraries, resources, and documentation, that users have access to

```
$ ls /usr
bin games include lib lib64 libexec local sbin share src
```

We found `/tmp` and it is showing the temporary files

```
$ ls /tmp
snap-private-tmp
systemd-private-92cc769f4f694c4280713f593dfde62e-ModemManager.service-fp9xMc
systemd-private-92cc769f4f694c4280713f593dfde62e-chrony.service-a0eiyr
systemd-private-92cc769f4f694c4280713f593dfde62e-fwupd.service-cAPlRrQ
systemd-private-92cc769f4f694c4280713f593dfde62e-polkit.service-yKKEzP
systemd-private-92cc769f4f694c4280713f593dfde62e-systemd-logind.service-Ur0eUR
systemd-private-92cc769f4f694c4280713f593dfde62e-systemd-resolved.service-eiGKD5
```

Step 3:

Use `find` and `ls -la` to locate every file under `/home` and `/etc/passwd`-related configuration, noting anything unexpected.

```
$ find /home -ls
 1684      4 drwxr-xr-x   5 root    root      4096 Aug  5 16:56 /home
 262145    4 drwxr-x---   3 azureuser azureuser 4096 Aug  5 16:42 /home/
azureuser
find: '/home/azureuser': Permission denied
 262324    4 drwxr-x---   3 student  student  4096 Aug  5 16:55 /home/
student
 262325    4 -rw-r--r--   1 student  student  3771 Mar 31 2024 /home/
student/.bashrc
 262326    4 -rw-r--r--   1 student  student   220 Mar 31 2024 /home/
student/.bash_logout
 262327    4 -rw-r--r--   1 student  student   807 Mar 31 2024 /home/
student/.profile
 262163    4 drwx-----   2 student  student  4096 Aug  5 16:55 /home/
student/.cache
 262166    0 -rw-r--r--   1 student  student    0 Aug  5 16:55 /home/
student/.cache/motd.legal-displayed
 262200    4 drwxr-x---   2 oldservice oldservice 4096 Aug  5 16:56 /hom
e/oldservice
find: '/home/oldservice': Permission denied
```

We then used `ls -la` to search for files under `/home` and found `oldservice` unusual and unexpected

```
$ pwd
/home/student
$ find /etc/passwd
/etc/passwd
$ ls -a
.  ..  .bash_logout  .bashrc  .cache  .profile
$ ls -la
total 24
drwxr-x--- 3 student student 4096 Aug  5 16:55 .
drwxr-xr-x 5 root    root    4096 Aug  5 16:56 ..
-rw-r--r-- 1 student student 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 student student 3771 Mar 31 2024 .bashrc
drwx----- 2 student student 4096 Aug  5 16:55 .cache
-rw-r--r-- 1 student student 807 Mar 31 2024 .profile
```

We continued are search and couldn't quite find anything.

```
$ ls -la /etc/passwd
-rw-r--r-- 1 root root 1883 Aug  5 16:56 /etc/passwd
```

We ran a different command to find the results shown below, and again the only thing that was unexpected was oldservice.

```
drwxr-xr-x 112 root root 4096 Aug  5 16:56 .
drwxr-xr-x  22 root root 4096 Aug  5 16:44 ..
-rw-----  1 root root    0 Aug  2 04:15 .pwd.lock
-rw-r--r--  1 root root  862 Aug  2 04:15 .resolv.conf.systemd-resolved.b
-rw-r--r--  1 root root  208 Aug  2 04:15 .updated
drwxr-xr-x  4 root root 4096 Aug  2 04:17 ModemManager
drwxr-xr-x  2 root root 4096 Aug  2 04:18 PackageKit
drwxr-xr-x  4 root root 4096 Aug  2 04:16 X11
-rw-r--r--  1 root root 3444 Jul  5 2023 adduser.conf
drwxr-xr-x  2 root root 4096 Aug  2 04:21 alternatives
drwxr-xr-x  2 root root 4096 Aug  2 04:17 apparmor
drwxr-xr-x  9 root root 4096 Aug  2 04:21 apparmor.d
drwxr-xr-x  3 root root 4096 Aug  2 04:18 appport
drwxr-xr-x  8 root root 4096 Aug  2 04:21 apt
-rw-r--r--  1 root root 2319 Mar 31 2024 bash.bashrc
-rw-r--r--  1 root root   45 Jan 24 2020 bash_completion
drwxr-xr-x  2 root root 4096 Aug  2 04:18 bash_completion.d
-rw-r--r--  1 root root  367 Aug  2 2022 bindresvport.blacklist

drwxr-xr-x  2 root root 4096 Aug  2 04:18 bash_completion.d
-rw-r--r--  1 root root  367 Aug  2 2022 bindresvport.blacklist
drwxr-xr-x  2 root root 4096 Apr 19 2024 binfmt.d
drwxr-xr-x  2 root root 4096 Aug  2 04:17 byobu
drwxr-xr-x  3 root root 4096 Aug  2 04:15 ca-certificates
-rw-r--r--  1 root root 6862 Aug  2 04:16 ca-certificates.conf
-rw-r--r--  1 root root 6288 Aug  2 04:16 ca-certificates.conf.dpkg-old
drwxr-xr-x  4 root root 4096 Aug  2 04:21 chrony
drwxr-xr-x  2 root root 4096 Aug  2 04:21 cifs-utils
drwxr-xr-x  5 root root 4096 Aug  2 04:18 cloud
drwxr-xr-x  2 root root 4096 Aug  2 04:18 console-setup
drwx----- 2 root root 4096 Apr 19 2024 credstore
drwx----- 2 root root 4096 Apr 19 2024 credstore.encrypted
drwxr-xr-x  2 root root 4096 Aug  2 04:17 cron.d
drwxr-xr-x  2 root root 4096 Aug  2 04:18 cron.daily
drwxr-xr-x  2 root root 4096 Aug  2 04:15 cron.hourly
drwxr-xr-x  2 root root 4096 Aug  2 04:15 cron.monthly
drwxr-xr-x  2 root root 4096 Aug  2 04:18 cron.weekly
```

Flagged:
/home/oldservice

Step 4:

List all running processes (ps aux) and all listening network services (ss -tulpn), and record them in your findings table.

```
$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.0  0.1 22484 14196 ?        Ss   16:44   0:04 /sbin/init
root             2  0.0  0.0      0     0 ?        S    16:44   0:00 [kthreadd]
root             3  0.0  0.0      0     0 ?        S    16:44   0:00 [pool_workqueue_release]
root             4  0.0  0.0      0     0 ?        I<   16:44   0:00 [kworker/R-rcu_gp]
root             5  0.0  0.0      0     0 ?        I<   16:44   0:00 [kworker/R-sync_wq]
root             6  0.0  0.0      0     0 ?        I<   16:44   0:00 [kworker/R-kvfree_rcu_reclaim]
root             7  0.0  0.0      0     0 ?        I<   16:44   0:00 [kworker/R-slub_flushwq]
root             8  0.0  0.0      0     0 ?        I<   16:44   0:00 [kworker/R-netns]
root            11  0.0  0.0      0     0 ?        I<   16:44   0:00 [kworker/0:0H-events_highpri]
root            13  0.0  0.0      0     0 ?        I<   16:44   0:00 [kworker/R-mm_percpu_wq]
root            14  0.0  0.0      0     0 ?        S    16:44   0:00 [ksoftirqd/0]
root            15  0.0  0.0      0     0 ?        I    16:44   0:00 [rcu_sched]
root            16  0.0  0.0      0     0 ?        S    16:44   0:00 [rcu_exp_par_gp_kthread_worker/0]
root            17  0.0  0.0      0     0 ?        S    16:44   0:00 [rcu_exp_gp_kthread_worker]
root            18  0.0  0.0      0     0 ?        S    16:44   0:00 [migration/0]
root            19  0.0  0.0      0     0 ?        S    16:44   0:00 [idle_inject/0]
```

Listening services include SSH port 22 and dns port 53.

Step 5:

Review currently installed packages and their versions (apt list --installed) and flag any obviously outdated or unnecessary software.

```
$ apt list --installed
Listing... Done
adduser/noble,now 3.137ubuntu1 all [installed,automatic]
apparmor/noble-updates,now 4.0.1really4.0.1-0ubuntu0.24.04.7 amd64 [installed,automatic]
appreport-core-dump-handler/noble-updates,now 2.28.2-0ubuntu0.1 all [installed,automatic]
appreport-symptoms/noble,now 0.25 all [installed,automatic]
apport/noble-updates,now 2.28.2-0ubuntu0.1 all [installed,automatic]
appstream/noble,now 1.0.2-1build6 amd64 [installed,automatic]
apt-utils/noble-updates,now 2.8.3 amd64 [installed,automatic]
apt/noble-updates,now 2.8.3 amd64 [installed,automatic]
azure-vm-utils/noble-updates,now 0.6.0-4~ubuntu0.24.04.1 amd64 [installed]
base-files/noble-updates,now 13ubuntu10.4 amd64 [installed]
base-passwd/noble,now 3.6.3build1 amd64 [installed,automatic]
bash-completion/noble,now 1:2.11-8 all [installed,automatic]
bash/noble,now 5.2.21-2ubuntu4 amd64 [installed]
bc/noble,now 1.07.1-3ubuntu4 amd64 [installed,automatic]
bcache-tools/noble,now 1.0.8-5build1 amd64 [installed,automatic]
bind9-dnsutils/noble-updates,noble-security,now 1:9.18.39-0ubuntu0.24.04.5 amd64 [installed,automatic]
```

Flagged software:
telnet/noble-updates,noble-security,now 0.17+2.5-3ubuntu4.2 all [installed,automatic]
ftp/noble,now 20230507-2build3 all [installed,automatic]

Summary

Category	Item	Finding	Risk
User/Home dir	oldservice	Present, unusual naming, permissions restricted	Flagged — verify shell/password Day 4
Listening service	:22, :53	SSH + local DNS only	None — expected
Package	telnet	Cleartext login protocol	High — remove
Package	ftp	Cleartext file transfer	High — remove
Package	netcat-openbsd	Dual-use diagnostic/pivot tool	Low — monitor

Day 4 Lab

Step 1:

Review all local user accounts (cat /etc/passwd) and identify any accounts that shouldn't exist or shouldn't have shell access

```
Last login: Thu Aug  6 19:44:51 2026 from 108.32.74.17
$ pwd
/home/student
$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
```

While not shown old-service, nobody, man

Step 2:

Audit sudo access (cat /etc/sudoers and /etc/sudoers.d/) and identify any overly broad privilege grants.

```
root@LinuxTargetVM:~# pwd
/root
root@LinuxTargetVM:~# cat /etc/sudoers
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
```

```
# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin  ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
```

```
root@LinuxTargetVM:~# ls /etc/sudoers.d
90-cloud-init-users  README  logreview  waagent
```

```
root@LinuxTargetVM:~# ls -la /etc/sudoers.d
. . . 90-cloud-init-users  README  logreview  waagent
```

```
root@LinuxTargetVM:~# less /etc/sudoers.d/90-cloud-init-users
```

```
# Created by cloud-init v. 20.1ubuntu1~24.04.1 on Wed, 03 Aug 2022 16:42:06 +0000

# User rules for azureuser
azureuser ALL=(ALL) NOPASSWD:ALL
~
~
~
~
~
~
~
(FND)
```

Step 3:

Find world-writable files and directories: `find / -xdev -type f -perm -0002 -ls` and determine which ones are a genuine risk.

Before

```
$ pwd
/home/student
$ sudo -i
[sudo] password for student:
root@LinuxTargetVM:~# find / -xdev -type f -perm -0002 -ls
 34563    0 -rw-rw-rw-    1 oldservice oldservice    0 Aug  7 13:57 /etc/old_report.log
   16     0 -rwxrwxrwx    1 root      root          0 Aug  7 13:56 /var/tmp/app_config.bak
  1474    0 -rwxrwxrwx    1 root      root          0 Aug  7 13:56 /tmp/backup_notes.txt
root@LinuxTargetVM:~#
```

All three are risks

Step 4:

Correct at least three permission or ownership issues you found using `chmod` / `chown`, documenting the before-and-after state for each.

Changing permission for app_config.bak

```
$ pwd
/home/student
$ sudo find / -xdev -type f -perm -0002 -ls
   16     0 -rwxrwxrwx    1 root      root          0 Aug  7 13:56 /var/tmp/app_config.bak
  1474    0 -rwxrwxrwx    1 root      root          0 Aug  7 13:56 /tmp/backup_notes.txt
$ chmod 700 app_config.bak
chmod: cannot access 'app_config.bak': No such file or directory
$ chmod 700 /var/tmp/app_config.bak
chmod: changing permissions of '/var/tmp/app_config.bak': Operation not permitted
$ sudo -i
root@LinuxTargetVM:~# chmod 700 /var/tmp/app_config.bak
root@LinuxTargetVM:~# sudo find / -xdev -type f -perm -0002 -ls
  1474    0 -rwxrwxrwx    1 root      root          0 Aug  7 13:56 /tmp/backup_notes.txt
root@LinuxTargetVM:~#
```

After:

Changing permission for old_report.log

```
root@LinuxTargetVM:~# chmod 444 /etc/old_report.log
root@LinuxTargetVM:~# find / -xdev -type f -perm -0002 -ls
   16     0 -rwxrwxrwx    1 root      root          0 Aug  7 13:56 /var/tmp/app_config.bak
  1474    0 -rwxrwxrwx    1 root      root          0 Aug  7 13:56 /tmp/backup_notes.txt
root@LinuxTargetVM:~#
```

Changing permission for back_notes.txt

```
root@LinuxTargetVM:~# chmod 740 /tmp/backup_notes.txt
root@LinuxTargetVM:~# find / -xdev -type f -perm -0002 -ls
root@LinuxTargetVM:~#
```

Step 5:

Apply least-privilege sudo rules for any account that needs elevated access to only the specific commands it requires.

```
# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
oldservice ALL=(ALL) NOPASSWD: ALL
```

Old service does not require a password at all at has elevated rights.

```
@includedir /etc/sudoers.d
# oldservice ALL=(ALL) NOPASSWD: ALL
```

Removed oldservice completely by commenting it out

Day 5 Lab

Step 1:

With your instructor's go-ahead, simulate an SSH brute-force attempt against the target from your jump host (a scripted series of failed login attempts) and observe it in the auth log: `tail -f /var/log/auth.log`

```
root@LinuxTargetVM:~# tail -f /var/log/auth.log
2026-08-07T17:42:53.989160+00:00 LinuxTargetVM sshd[53244]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh r
user= rhost=103.200.23.154 user=root
2026-08-07T17:42:55.184566+00:00 LinuxTargetVM sshd[53244]: Failed password for root from 103.200.23.154 port 53606 ssh2
2026-08-07T17:42:56.356668+00:00 LinuxTargetVM sshd[53244]: Received disconnect from 103.200.23.154 port 53606:11: Bye Bye [preauth]
2026-08-07T17:42:56.356866+00:00 LinuxTargetVM sshd[53244]: Disconnected from authenticating user root 103.200.23.154 port 53606 [preaut
h]
2026-08-07T17:43:31.432066+00:00 LinuxTargetVM sshd[53246]: Invalid user mc from 190.181.27.37 port 47820
2026-08-07T17:43:31.435379+00:00 LinuxTargetVM sshd[53246]: pam_unix(sshd:auth): check pass; user unknown
2026-08-07T17:43:31.435546+00:00 LinuxTargetVM sshd[53246]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh r
user= rhost=190.181.27.37
2026-08-07T17:43:33.514318+00:00 LinuxTargetVM sshd[53246]: Failed password for invalid user mc from 190.181.27.37 port 47820 ssh2
2026-08-07T17:43:35.610391+00:00 LinuxTargetVM sshd[53246]: Received disconnect from 190.181.27.37 port 47820:11: Bye Bye [preauth]
2026-08-07T17:43:35.610589+00:00 LinuxTargetVM sshd[53246]: Disconnected from invalid user mc 190.181.27.37 port 47820 [preauth]
2026-08-07T17:45:01.165359+00:00 LinuxTargetVM CRON[53252]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-08-07T17:45:01.169173+00:00 LinuxTargetVM CRON[53252]: pam_unix(cron:session): session closed for user root
2026-08-07T17:45:04.845444+00:00 LinuxTargetVM sshd[53255]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh r
user= rhost=92.118.39.50 user=root
2026-08-07T17:45:07.024527+00:00 LinuxTargetVM sshd[53255]: Failed password for root from 92.118.39.50 port 37824 ssh2
2026-08-07T17:45:09.631367+00:00 LinuxTargetVM sshd[53255]: Connection closed by authenticating user root 92.118.39.50 port 37824 [preau
th]
2026-08-07T17:45:19.290602+00:00 LinuxTargetVM sshd[53261]: Invalid user dev from 103.200.23.154 port 45678
```

Live log tracking

Step 2:

Install and configure fail2ban to automatically block repeated failed SSH login attempts, then re-run your brute-force simulation to confirm it's blocked.

```
Last login: Fri Aug 7 17:40:03 2026 from 20.120.96.152
$ pwd
/home/student
$ sudo apt install fail2ban
[sudo] password for student:
Sorry, try again.
[sudo] password for student:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pyasyncore python3-pyinotify whois
Suggested packages:
```

This is the proof of install for fail2ban.

```
root@LinuxTargetVM:~# sudo systemctl enable --now fail2ban
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
root@LinuxTargetVM:~# tail -f /var/log/auth.log
2026-08-07T17:48:57.226711+00:00 LinuxTargetVM sshd[53368]: Received disconnect from 190.181.27.37 port 34490:11: Bye Bye [preauth]
2026-08-07T17:48:57.226899+00:00 LinuxTargetVM sshd[53368]: Disconnected from authenticating user root 190.181.27.37 port 34490 [preauth]
2026-08-07T17:49:08.529040+00:00 LinuxTargetVM sudo: pam_unix(sudo:session): session closed for user root
2026-08-07T17:49:47.677537+00:00 LinuxTargetVM sshd[53654]: Connection closed by 186.158.230.16 port 33877
2026-08-07T17:49:49.867132+00:00 LinuxTargetVM sshd[53655]: Connection closed by 186.158.230.16 port 20319
2026-08-07T17:50:43.477135+00:00 LinuxTargetVM sshd[53663]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=92.118.39.50 user=root
2026-08-07T17:50:45.861863+00:00 LinuxTargetVM sshd[53663]: Failed password for root from 92.118.39.50 port 44168 ssh2
2026-08-07T17:51:23.901737+00:00 LinuxTargetVM sudo: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/systemctl enable --now fail2ban
2026-08-07T17:51:23.902843+00:00 LinuxTargetVM sudo: pam_unix(sudo:session): session opened for user root(uid=0) by student(uid=0)
2026-08-07T17:51:25.602259+00:00 LinuxTargetVM sudo: pam_unix(sudo:session): session closed for user root
```

Confirm blocked ssh attempts after enabling fail2ban

Step 3:

Disable direct root SSH login and password-based SSH authentication in favor of key-based authentication.

```
# ForceCommand cvs server
ChallengeResponseAuthentication no
PasswordAuthentication yes
PermitRootLogin yes
```

Before disabling ssh login and password-based authentication in favor of key based authentication

```
ChallengeResponseAuthentication no
PasswordAuthentication no
PermitRootLogin no
```

After disabling ssh login and password-based authentication of key based authentication

```
#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
```

Enabling key based authentication.

```
root@LinuxTargetVM:~# sudo nano /etc/ssh/sshd_config
root@LinuxTargetVM:~# sudo systemctl restart ssh
root@LinuxTargetVM:~# pwd
/root
root@LinuxTargetVM:~# ssh student@
```

After disabling restarting the ssh service

```
2026-08-07T18:19:31.480486+00:00 LinuxTargetVM sshd[50380]: Received disconnect from 73.199.75.168 port 1824:11: disconnected by user
2026-08-07T18:19:31.480670+00:00 LinuxTargetVM sshd[50380]: Disconnected from user student 73.199.75.168 port 1824
2026-08-07T18:19:31.481489+00:00 LinuxTargetVM sshd[50320]: pam_unix(sshd:session): session closed for user student
2026-08-07T18:19:31.487405+00:00 LinuxTargetVM sudo: pam_unix(sudo-i:session): session closed for user root
2026-08-07T18:19:31.489953+00:00 LinuxTargetVM systemd-logind[753]: Session 375 logged out. Waiting for processes to exit.
2026-08-07T18:19:31.498781+00:00 LinuxTargetVM systemd-logind[753]: Removed session 375.
2026-08-07T18:20:02.398444+00:00 LinuxTargetVM sshd[58649]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=73.199.75.168 user=root
2026-08-07T18:20:04.597079+00:00 LinuxTargetVM sshd[58649]: Failed password for root from 73.199.75.168 port 15698 ssh2
2026-08-07T18:20:09.235951+00:00 LinuxTargetVM sshd[58657]: Connection closed by 181.42.134.184 port 30211
```

Proof that disabled root logging.

Step 4:

Apply available system updates and patches: `apt update` & `apt upgrade`, and record what was patched.

```
root@LinuxTargetVM:~# pwd
/root
root@LinuxTargetVM:~# sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1184 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [281 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [180 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1682 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [335 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5744 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-security/main amd64 Packages [927 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-security/main Translation-en [201 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1320 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu noble-security/restricted Translation-en [302 kB]
Fetched 7320 kB in 2s (3671 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@LinuxTargetVM:~# |
```

Use `apt update` to refresh packages and version

```
root@LinuxTargetVM:~# sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  linux-azure-6.17.0-1022-azure linux-image-6.17.0-1022-azure linux-modules-6.17.0-1022-azure linux-tools-6.17.0-1022-azure
The following upgrades have been deferred due to phasing:
  apport apport-core-dump-handler python3-apport python3-problem-report
The following packages will be upgraded:
  linux-azure linux-cloud-tools-azure linux-headers-azure linux-image-azure linux-tools-azure
5 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
Need to get 72.8 MB of archives.
After this operation, 187 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-modules-6.17.0-1022-azure amd64 6.17.0-1022.22 [36.7 MB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-image-6.17.0-1022-azure amd64 6.17.0-1022.22 [15.3 MB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-azure amd64 6.17.0-1022.22 [1759 B]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-image-azure amd64 6.17.0-1022.22 [2436 B]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-6.17.0-1022 all 6.17.0-1022.22 [14.4 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-6.17.0-1022-azure amd64 6.17.0-1022.22 [3515 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-azure amd64 6.17.0-1022.22 [2226 B]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-azure-6.17.0-1022 amd64 6.17.0-1022.22 [2738 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.17.0-1022-azure amd64 6.17.0-1022.22 [1632 B]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-azure amd64 6.17.0-1022.22 [2338 B]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-azure-6.17.0-1022 amd64 6.17.0-1022.22 [54.6 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-cloud-tools-6.17.0-1022-azure amd64 6.17.0-1022.22 [1616 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-cloud-tools-azure amd64 6.17.0-1022.22 [2352 B]
Fetched 72.8 MB in 5s (14.8 MB/s)
Selecting previously unselected package linux-modules-6.17.0-1022-azure.
(Reading database ... 69985 files and directories currently installed.)
Preparing to unpack .../60-linux-modules-6.17.0-1022-azure_6.17.0-1022.22_amd64.deb ...
Unpacking linux-modules-6.17.0-1022-azure (6.17.0-1022.22) ...
Selecting previously unselected package linux-image-6.17.0-1022-azure.
Preparing to unpack .../61-linux-image-6.17.0-1022-azure_6.17.0-1022.22_amd64.deb ...
Unpacking linux-image-6.17.0-1022-azure (6.17.0-1022.22) ...
Preparing to unpack .../62-linux-azure_6.17.0-1022.22_amd64.deb ...
Unpacking linux-azure (6.17.0-1022.22) over (6.17.0-1021.21-24.04.1) ...
Preparing to unpack .../63-linux-image-azure_6.17.0-1022.22_amd64.deb ...
Unpacking linux-image-azure (6.17.0-1022.22) over (6.17.0-1021.21-24.04.1) ...
Selecting previously unselected package linux-headers-6.17.0-1022-azure.
Preparing to unpack .../64-linux-headers-6.17.0-1022-azure_6.17.0-1022.22_amd64.deb ...
Unpacking linux-headers-6.17.0-1022-azure (6.17.0-1022.22) ...
Preparing to unpack .../65-linux-headers-6.17.0-1022-azure_6.17.0-1022.22_amd64.deb ...
Unpacking linux-headers-6.17.0-1022-azure (6.17.0-1022.22) over (6.17.0-1021.21-24.04.1) ...
Preparing to unpack .../66-linux-headers-azure_6.17.0-1022.22_amd64.deb ...
Unpacking linux-headers-azure (6.17.0-1022.22) over (6.17.0-1021.21-24.04.1) ...
Preparing to unpack .../67-linux-azure-6.17.0-1022-azure_6.17.0-1022.22_amd64.deb ...
Unpacking linux-azure-6.17.0-1022-azure (6.17.0-1022.22) over (6.17.0-1021.21-24.04.1) ...
```


Use apt upgrade to patch and upgraded needed packages

```
Pending kernel upgrade!
Running kernel version:
  6.17.0-1021-azure
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.17.0-1022-azure.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

Upgrade confirmed

```
root@LinuxTargetVM:~# pwd
/root
root@LinuxTargetVM:~# systemctl list-units --type=service --state=running
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
chrony.service	loaded	active	running	chrony, an NTP client/server
cron.service	loaded	active	running	Regular background program processing daemon
dbus.service	loaded	active	running	D-Bus System Message Bus
fail2ban.service	loaded	active	running	Fail2Ban Service
fwupd.service	loaded	active	running	Firmware update daemon
getty@tty1.service	loaded	active	running	Getty on tty1
hv-kvp-daemon.service	loaded	active	running	Hyper-V KVP Protocol Daemon
inetutils-inetd.service	loaded	active	running	GNU Network Utilities internet superserver
ModemManager.service	loaded	active	running	Modem Manager
multipathd.service	loaded	active	running	Device-Mapper Multipath Device Controller

Telnet is running unsecure on port 23

```
root@LinuxTargetVM:~# pwd
/root
root@LinuxTargetVM:~# systemctl status inetutils-inetd.service
```

```
● inetutils-inetd.service - GNU Network Utilities internet superserver
   Loaded: loaded (/usr/lib/systemd/system/inetutils-inetd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-08-07 14:16:41 UTC; 4h 16min ago
     Docs: man:inetutils-inetd(8)
           https://www.gnu.org/software/inetutils/manual/
   Main PID: 51291 (inetutils-inetd)
     Tasks: 1 (limit: 9432)
    Memory: 208.0K (peak: 1.4M)
       CPU: 13ms
```

Showing the current status of the telnet service.

```
root@LinuxTargetVM:~# systemctl stop inetutils-inetd && systemctl disable inetutils-inetd
Synchronizing state of inetutils-inetd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable inetutils-inetd
Removed "/etc/systemd/system/multi-user.target.wants/inetutils-inetd.service".
Removed "/etc/systemd/system/inetd.service".
root@LinuxTargetVM:~#
```

Remove/disable telnet (hardened system)

```
lines 2-14/14 (END)
^C
root@LinuxTargetVM:~# systemctl status inetutils-inetd.service
o inetutils-inetd.service - GNU Network Utilities internet superserver
   Loaded: loaded (/usr/lib/systemd/system/inetutils-inetd.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: man:inetutils-inetd(8)
           https://www.gnu.org/software/inetutils/manual/

Aug 07 14:13:26 LinuxTargetVM systemd[1]: Started inetutils-inetd.service - GNU Network Utilities internet superserv
Aug 07 14:13:26 LinuxTargetVM inetd[51216]: telnet/tcp: No such user 'telnetd', service ignored
Aug 07 14:16:41 LinuxTargetVM systemd[1]: Stopping inetutils-inetd.service - GNU Network Utilities internet superserv
Aug 07 14:16:41 LinuxTargetVM systemd[1]: inetutils-inetd.service: Deactivated successfully.
Aug 07 14:16:41 LinuxTargetVM systemd[1]: Stopped inetutils-inetd.service - GNU Network Utilities internet superserv
```

Proof that telnet was removed/disabled.

Reflection Questions:

1. Having telnet as a running service was a major security weakness due to its insecure nature. Telnet sends credentials across a network without encryption paired with the running of “oldservice”, this system was weak. The overall permissive oldservice account having NOPASSWD sudo privileges was marked as a critical risk.
2. A fix that closes a specific hole only patches one vulnerability while a fix that changes the host's overall posture is more strategic and proactive. An example of a fix that closes a specific hole is correcting permissions and ownership of the three world writable files and directories. It only fixed one issue at a time within the system. While installing and configuring fail2ban was a more strategic and proactive way of fixing ssh root logins.
3. If this was someone else host, an updated inventory of running services, user/group list, and logs on why uncommon permissions exist. A trail of every change and modification is included in this report through screenshot showing every command that was ran.